

SIH 2026 GRAND
FINALE PITCH

DURATION: EXACTLY 3:00 (180
SECONDS)

PROBLEM STATEMENT: CIRCULAR FOOD
WASTE REDUCTION

Loife — Good Food. Shared Life.

Enterprise AI-Powered Food Waste Reduction, Safety Clearance & Automated Redistribution Platform

Target Target

Smart India Hackathon 2026

Pacing Strategy

~135 words/minute (Fluent,
Convincing)

Demo Credentials

admin@reserveai.com /
Admin@1234

Primary Impact

Zero Hunger (SDG 2) +
Climate Action (SDG 13)

Scene 1: The National Crisis & The Broken Loop

0:00 - 0:25 (25s)

Screen Action: Open <http://localhost:3000/> (Landing Page). Slowly scroll past the glowing header, showing the tagline "Good Food. Shared Life.", the animated ambient lights, and pause over the community stories section.

SPOKEN VOICEOVER

"Every year, India wastes over **68 million tonnes of edible food** — worth nearly 89,000 crore rupees — while over 190 million citizens sleep malnourished. In universities, IT campuses, and hospital mess halls, culinary teams cook in bulk with zero visibility into actual demand. Surplus is discarded to landfills, generating dangerous methane and squandering thousands of litres of water per kilogram.

The core issue isn't goodwill; it is **lack of closed-loop predictive technology**. Today, we present **Loife** — India's first end-to-end circular food ecosystem powered by Artificial Intelligence."

★ JURY HOOK: Immediate macroeconomic and humanitarian framing directly aligned with national hackathon mandates.

Scene 2: Executive Overview & Scientific ESG Accounting

0:25 - 0:50 (25s)

Screen Action: Click "**Sign In**" → Click the "**Admin**" quick-demo pill → Submit to land on **Executive Overview**. Hover over the KPI cards: 14.25 MT Rescued, 37.03 t CO₂e Avoided, 8.7M Litres Water Saved. Click "**Export ESG Report**" to show the instant JSON audit download.

SPOKEN VOICEOVER

"Logging in as an Executive Director, Loife provides an institutional command center. Every kilogram of rescued food is scientifically audited using peer-reviewed **Poore & Nemecek (2018) Life Cycle Assessment** standards.

Here, our campus dining cluster has diverted **14.25 metric tonnes** of food, avoided **37 tonnes of CO₂ equivalent**, conserved **8.7 million litres of virtual embedded water**, and returned over **15.6 lakh rupees** in operational efficiency. With one click, institutions generate certified ESG and Scope 3 compliance reports ready for audit."

★ JURY HOOK: Proves mathematical rigor. Not arbitrary numbers, but peer-reviewed LCA environmental science.

Scene 3: Kitchen Operations & LightGBM Demand Forecasting

0:50 - 1:20 (30s)

Screen Action: In the sidebar, click "**Kitchen Management**". Point cursor to the **10 animated glowing batch circles** with radiant motes of light floating randomly. Hover over 2 circles to show batch expiry tooltips. Click "**Run AI Forecast**" → Modal pops up → Click "**Generate & Persist**".

SPOKEN VOICEOVER

"The best way to eliminate waste is to prevent it at the source. In the Kitchen Operations cockpit, our **LightGBM predictive engine** forecasts meal-level footfall by factoring historical attendance trends, academic calendar surges, and inventory on hand.

Notice our **real-time Shelf-Life Radar** — these 10 animated alert bubbles track perishable batches. Green radiant motes represent optimal batches, while amber bubbles highlight items expiring within 5 hours. Chef teams receive automated recommendations to throttle prep batches by exact kilograms or declare surplus with zero guesswork."

★ JURY HOOK: Shows active prevention. Most solutions only redistribute waste; Loife prevents it before cooking starts.

Scene 4: Computer Vision Food Quality & Multi-Factor Safety

1:20 - 1:45 (25s)

Screen Action: Navigate to "**CV Freshness Scan**". Click the "**Fresh Produce**" benchmark preset. Point to the optical freshness score (91.3% Fresh), shelf-life projection (4.9 days), and IoT temperature clearance. Click "**Approve & Sign-Off**".

SPOKEN VOICEOVER

"Safety is paramount in food donation. Before any surplus can leave a facility, it undergoes Loife's **Computer Vision Optical Clearance**.

Our convolutional neural vision model analyzes surface pigmentation, cellular degradation, and moisture loss, returning a 91.3% freshness rating and residual shelf-life prediction. Combined with IoT cold-chain sensor clearance and mandatory human inspector sign-off, we guarantee strict **FSSAI hygiene compliance** before a single portion is dispatched."

★ JURY HOOK: Solves the #1 legal/health liability holding back institutional dining from donating cooked surplus.

Scene 5: Autonomous Multi-Factor NGO Matchmaking

1:45 – 2:15 (30s)

Screen Action: Navigate to "**Redistribution Hub**". Click on **Paneer Butter Masala (35 kg, ~85 meals)**. Point to the right-hand AI Compatibility ranking showing *Feeding India by Zomato (77.5%)* and *Roti Bank (75.1%)*. Click "**Dispatch**". Show the instant green confirmation banner: "*Dispatch Scheduled*".

SPOKEN VOICEOVER

"When surplus is declared, Loife's **Autonomous Matching Engine** activates. Instead of endless manual phone calls, our algorithm pairs lots using a 4-factor scoring model: distance radius, NGO shelter capacity, cold storage capability, and verified historical reliability.

As we select this 35kg lot, our engine ranks **Feeding India by Zomato** as the optimal match with a 77.5% compatibility score. Clicking 'Dispatch' instantly locks the allocation, reserves the shelter capacity, and schedules logistics in under 3 seconds."

★ JURY HOOK: Shows real-time autonomous routing replacing manual delays that cause food to spoil before pickup.

Scene 6: Smart EV Logistics & Cryptographic Proof-of-Delivery

2:15 – 2:40 (25s)

Screen Action: In the sidebar, click "**Logistics & Routes**". Show the active EV route code RT-DELHI-NORTH-01, vehicle telemetry (3.4°C cold-chain reefer, driver Rajesh Kumar). Scroll down to the Digital Proof of Delivery card. Enter OTP 8492 and click "**Verify & Complete Delivery**".

SPOKEN VOICEOVER

"In the Logistics command dashboard, a greedy heuristic algorithm aggregates multiple kitchen pick-ups into single electric vehicle runs.

Notice our live reefer telemetry maintaining an unbroken **3.4°C cold chain**. At the recipient hub, the NGO representative provides a secure **4-digit verification OTP**. Once entered, the transaction is cryptographically stamped into the audit ledger, certifying temperature, quantity, and verified handover with zero fraud."

★ JURY HOOK: Complete trace-and-track accountability that eliminates diversion and ghost-deliveries.

Scene 7: Scalability, Impact & Final Call to Action

2:40 – 3:00 (20s)

Screen Action: Switch to the **Sustainability & ESG Dashboard** showing the Category Savings chart and the Scope 3 Audit Certificate, or back to the high-impact Landing Page footer.

SPOKEN VOICEOVER

"Loife is built on a scalable FastAPI and React architecture designed for municipal multi-tenant rollouts across India's thousands of campuses, airports, and corporate clusters.

By uniting predictive AI, computer vision safety, and transparent logistics, we transform food waste from an environmental catastrophe into community nourishment. Because with Loife — **Good Food is Shared Life**. Thank you."

★ **CLOSING NOTE:** Resonant finish connecting cutting-edge engineering with tangible societal impact.

Executive Presentation Master Cheat-Sheet

Total Runtime: 180s

Timestamp	Screen / Navigation	Action to Execute on Screen	Core Pitch Angle
0:00 – 0:25	Landing Page (/)	Scroll past golden lights & community stories	The 68M tonne Indian food crisis
0:25 – 0:50	Executive Overview	Click Admin Demo login → Hover on 14.25 MT KPI card	LCA-certified ESG Scope 3 compliance
0:50 – 1:20	Kitchen Management	Hover on 10 glowing batch bubbles → "Run AI Forecast"	Predictive prevention at source
1:20 – 1:45	CV Freshness Scan	Click "Fresh Produce" preset → Click "Approve & Sign-Off"	Optical ML freshness & FSSAI safety
1:45 – 2:15	Redistribution Hub	Select surplus lot → Click "Dispatch" to Feeding India	Autonomous 4-factor matchmaking
2:15 – 2:40	Logistics & Fleet	Show 3.4°C reefer telemetry → Verify OTP (8492)	Unbroken cold-chain & proof of delivery
2:40 – 3:00	Sustainability / ESG	Show Scope 3 Certificate modal → Concluding call to action	National scalability & UN SDGs

Architecture & Innovation Stack (For Technical Q&A)

Full-Stack Enterprise Grade

Predictive AI & CV Engines

LightGBM footfall models, Convolutional Neural optical freshness classifier (FSSAI benchmarked), Greedy heuristic routing.

Backend & Data Pipeline

FastAPI async REST architecture, SQLAlchemy ORM, SQLite/PostgreSQL, WebSocket telemetry, JWT session protection.

Frontend & UX System

React 18, TypeScript, Tailwind CSS, Recharts analytics, ambient particle physics, full mobile/desktop responsiveness.